

Assignment 4

Part 1

Primary key: sessionID

Functional Dependencies:

SessionID have dependents SessionDate, SessionTime, StudentID, TutorID, CourseID, RoomID, SessionType, hourlyRate, DurationMinutes

StudentID have dependents StudentName, StudentEmail, Major

TutorID have dependents TutorName, TutorEmail

courseID have dependents CoursTitle, Department

RoomID have dependents RoomBuilding, RoomCapacity

Anomalies:

Insertion Anomaly: You cannot add a new student unless they are assigned to a tutoring session.

Update Anomaly: If a tutor changes their email, you must update every row where that tutor appears.

Deletion Anomaly: If a tutoring session is deleted, important information could also be lost.

Part 2:

Student (StudentID PK, StudentName, StudentEmail, Major)

Tutor (TutorID PK, TutorName, TutorEmail)

Course(CourseID PK), CourseTitle, Department)

Room (RoomID PK, RoomBuilding, RoomCapacity)

Session (SessionID PK, SessionDate, SessionTime, StudentID FK, TutorID FK, CourseID FK, RoomID FK, SessionType, HourlyRate, DurationMinutes)

Pictures of query results:

```
mysql> SELECT
->     S.StudentName,
->     T.TutorName,
->     C.CourseTitle,
->     R.RoomID
-> FROM SESSION SE
-> JOIN STUDENT S ON SE.StudentID = S.StudentID
-> JOIN TUTOR T ON SE.TutorID = T.TutorID
-> JOIN COURSE C ON SE.CourseID = C.CourseID
-> JOIN ROOM R ON SE.RoomID = R.RoomID;
```

StudentName	TutorName	CourseTitle	RoomID
Alice Johnson	John Carter	Database Systems	301
Brian Smith	Sarah Kim	Calculus I	302
Cathy Lee	Michael Chen	Organic Chemistry	303
David Brown	Lisa White	Physics I	304
Emma Wilson	James Hall	Programming Fundamentals	305

5 rows in set (0.178 sec)

```
mysql> SELECT *
-> FROM SESSION
-> WHERE StudentID = 1;
```

SessionID	SessionDate	SessionTime	StudentID	TutorID	CourseID	RoomID	SessionType	HourlyRate	DurationMinutes
1	2026-05-01	10:00:00	1	101	201	301	Individual	25.00	60

1 row in set (0.114 sec)

```
mysql> SELECT *
-> FROM SESSION
-> WHERE TutorID = 101;
```

SessionID	SessionDate	SessionTime	StudentID	TutorID	CourseID	RoomID	SessionType	HourlyRate	DurationM inutes
1	2026-05-01	10:00:00	1	101	201	301	Individual	25.00	60

```
1 row in set (0.016 sec)
```

```
mysql> SELECT
-> CourseID,
-> COUNT(*) AS TotalSessions
-> FROM SESSION
-> GROUP BY CourseID;
```

CourseID	TotalSessions
201	1
202	1
203	1
204	1
205	1

```
5 rows in set (0.165 sec)
```

```
mysql> SELECT
->     RoomID,
->     COUNT(*) AS TotalSessions
-> FROM SESSION
-> GROUP BY RoomID;
```

RoomID	TotalSessions
301	1
302	1
303	1
304	1
305	1

5 rows in set (0.017 sec)